

XRPL Validator — Daily Engineering Report • 2026-09-13

Compiled retrospectively on 2026-09-23 from the git history. Scope: XRPL work only — the git history of [xrpl-validator](#) and the XRPL Rust SDK, the XRP Ledger itself (amendments) and XRPL research.

Summary

- **11 commits** (xrpl-validator 11)
- **11 engine findings fixed** (F267–F277)

Engine findings fixed

- **fix(xrpl-ledger): F267 — brokered NFT accept must cover the ask after the broker fee** (tecINSUFFICIENT_PAYMENT) (e094350, xrpl-validator) rippled 3.3.0 NFTTokenAcceptOffer::preclaim, brokered mode (:95-142): same token, same asset, no owner loop, so.Amount > bo.Amount -> tecINSUFFICIENT_PAYMENT; then, with an explicit NFTTokenBrokerFee, the fee must be the buy asset, below the buy amount, and the ask must still be covered after it. Our preclaim had none of the amount checks.
- **fix(xrpl-ledger): F268 — an out-limited leg-B book partial on the XRP bridge ceils the mid-leg strictly** (123e9a7, xrpl-validator) rippled sizes an OUT-limited partial fill of a book maker with limitStepOut → Toffer::limitOut → Quality::ceilOutStrict (Offer.h:216-221): a true ceiling of the exact product to whole drops. Our bridge's b_unprice used mul_round16_up, the lossy legacy mulRound, which discards everything below the 17th digit before its carry.
- **fix(xrpl-ledger): F270 — a book leg enters the single-strand quality function fee-free** (9f9f231, xrpl-validator) BookOfferCrossingStep::adjustQualityWithFees (BookStep.cpp:512-540) returns the offer quality untouched for OfferType::Clob — "Offer x-ing does not charge a transfer fee ... single path AMM's offer quality is not constant" — so only a single-path POOL leg carries trIn into the strand's QualityFunction; a book tip enters fee-free, while the limitQuality the function is solved against stays the fee-inflated one. F224 ...
- **fix(xrpl-ledger): F269 — a taker frozen on its own account cannot cross; the strand is never built** (1eb4a78, xrpl-validator) rippled's strand builder runs checkFreeze(view, src, dst) on the delivery step with dst = the taker (StepChecks.h; DirectStepI::check for an IOU, XRPEndpointStep::check with src = xrpAccount for XRP). It returns terNO_LINE when the taker's OWN AccountRoot carries lsfGlobalFreeze, or, for an IOU delivery, when the taker's line to the issuer carries the taker-side freeze bit or either deep-freeze bit. toStrand ...
- **fix(xrpl-ledger): F271 — the reverse-pass sizing ladder escalates past a rung that yields nothing** (c748500, xrpl-validator) size_book_hop measures a hop by granting 1e6 / 1e12 / 1e18 of the in-asset and reading what came out. A rung that yields NOTHING is not "bound by liquidity", it is below the hop's quantum: a pool priced at ~2.8e6 FACE per drop turns 1e6 FACE into 0.36 drops, which floors to zero. consumed == 0 < granted then read as a liquidity bound, the hop reported 0, the hop before it got the UNBOUNDED cap and consumed its ...

- **fix(xrpl-ledger): F272 — an in-limited forward pass floors the bridged mid-leg; zero drops is a dry bridge** (a7c12f5, xrpl-validator) When the whole drop the OUT side ceiled to costs more than the taker has left, rippled's forward pass is in-limited and leg A yields `floor(in / rate)` drops (F93's `limitStepIn` → `ceilInStrict(..., roundUp = false)`; a pool's `swapAssetIn` output rounds down the same way). We clamped `gets_in` to the remainder while keeping the ceiled drop — buying a drop the taker could not pay for.
- **fix(xrpl-ledger): F273 — the offer walk counts each pool slice as a flow iteration and stops at thirty** (617c445, xrpl-validator) `AMMLiquidity::getOffer` answers nullopt once `ammContext.maxItersReached()` (`kMaxIterations = 30`); the counter advances once per flow iteration whose winning pass used the pool (`setAMMUsed` → `update`). The offer walk IS the flow, so every slice it takes is one such iteration. Finding 107 gated only `discover`, which the walk calls once — the 31st slice and every one after it went through, and the single-path turn ...
- **fix(xrpl-ledger): F274 — a FillOrKill sell is judged on rippled's per-iteration sixteen-digit fold** (d48c66a, xrpl-validator) rippled's `remainingIn = *sendMax - sum(savedIns)` (`StrandFlow.h:745`) is the sixteen-digit sorted-multiset fold of the ITERATIONS' ins — one entry per book level or pool slice — and a FillOrKill sell is judged on that (:829-841: `remainingIn` must be zero or the flow returns `tecPATH_PARTIAL` and the offer is `tecKILLED`). The direct walk folded per FILL for Finding 236's rested amount, skipped that fold when its running ...
- **fix(xrpl-ledger): F275 — a "became unfunded" reap lives in its trial's sandbox and dies with a rejected trial** (88794e2, xrpl-validator) rippled's `FlowOfferStream` deletes an offer whose owner's funds are zero NOW but were not in the pristine view — "became unfunded" — in the strand's sandbox; the removal persists only if that strand wins the iteration and its sandbox is applied. A rejected trial ("Path rejected by limitQuality") is discarded, removal included. Only a "found unfunded", expired or domain removal is `permRmOffer`ed for good ...
- **fix(xrpl-ledger): F276 — an offer rate whose quotient reaches 2^63 wraps through Number's signed mantissa** (03b96c8, xrpl-validator) `divide` hands `muldiv(numVal, 1e17, denVal) + 5` to `STAmount(asset, uint64, exponent, negative)`, whose `canonicalize()` builds `Number(negative, value_, offset_)` — and `Number`'s mantissa is a signed 64-bit. A quotient at or above 2^63 wraps negative; the magnitude that survives is 2^64 - v, and `getRate` files it with the sign dropped (`STAmount.cpp canonicalize / getRate`, rippled 3.3.0).
- **fix(xrpl-ledger): F277 — a "became tiny" or dust reap is the strand's own; only a found-tiny one is permanent** (5d804cc, xrpl-validator) rippled's `FlowOfferStream` splits the reduced-quality (tiny) removal the way it splits the unfunded one (`OfferStream.cpp:308-332`): "Removing tiny offer due to reduced quality" is `permRmOffer` when the owner's funds are what the pristine view held, "Removing tiny offer that BECAME tiny due to reduced quality" lives in the strand's sandbox otherwise — discarded with a rejected trial or a killed transaction. Four of ...

Amendment watch

Read from the XRP Ledger: the Amendments object as of the last ledger of the day, and the day's EnableAmendment pseudo-transactions.

No amendment held a majority.

XRPL research

Automated research — candidate amendment ideas

Ideas generated and prior-art-checked by the XRPL Amendment Board, an automated research tool (read-only board: iceland.halcyon-names.io). They are exploration leads, not proposals.

- **BLS Aggregate Signatures for Signer Lists** (custody) Add an optional sfMPCPublicKey (STBlob, BLS12-381 compressed point) to the SignerList object, settable via a new tfMPCMode flag on SignerListSet in pl
- **On-Ledger NFT Rental Lease Object** (gaming) Add a new owner-reserved ledger object NFTTokenLease (ltNFTOKEN_LEASE, keylet::nftLease(nftID)) storing sfNFTTokenID, sfLessee, sfLeaseExpiration (ledge
- **AMM-Seeded Exit Liquidity for Fractional NFTs** (nft) Extend g12's NFTTokenFractionalize transaction with a tfSeedAMM flag: after locking the NFTToken into the NFTTokenVault and minting the fixed-supply MPTo

Appendix — commits

Time	Repo	Commit	Subject
00:28	xrpl-validator	e094350	fix(xrpl-ledger): F267 — brokered NFT accept must cover the ask after the broker fee (tecINSUFFICIENT_PAYMENT)
08:22	xrpl-validator	123e9a7	fix(xrpl-ledger): F268 — an out-limited leg-B book partial on the XRP bridge ceils the mid-leg strictly
08:42	xrpl-validator	9f9f231	fix(xrpl-ledger): F270 — a book leg enters the single-strand quality function fee-free
08:42	xrpl-validator	1eb4a78	fix(xrpl-ledger): F269 — a taker frozen on its own account cannot cross; the strand is never built
08:49	xrpl-validator	c748500	fix(xrpl-ledger): F271 — the reverse-pass sizing ladder escalates past a rung that yields nothing
09:04	xrpl-validator	a7c12f5	fix(xrpl-ledger): F272 — an in-limited forward pass floors the bridged mid-leg; zero drops is a dry bridge
09:13	xrpl-validator	617c445	fix(xrpl-ledger): F273 — the offer walk counts each pool slice as a flow iteration and stops at thirty
11:36	xrpl-validator	d48c66a	fix(xrpl-ledger): F274 — a FillOrKill sell is judged on rippled's per-iteration sixteen-digit fold
11:50	xrpl-validator	88794e2	fix(xrpl-ledger): F275 — a "became unfunded" reap lives in its trial's sandbox and dies with a rejected trial
17:48	xrpl-validator	03b96c8	fix(xrpl-ledger): F276 — an offer rate whose quotient reaches 2^63 wraps through Number's signed mantissa
18:14	xrpl-validator	5d804cc	fix(xrpl-ledger): F277 — a "became tiny" or dust reap is the strand's own; only a found-tiny one is permanent